

Enterprise Dashboard Enhancement

Customizable reporting and automated PDF scheduling to reduce churn and increase retention for high-value enterprise accounts.

FOR: EXECUTIVE LEADERSHIP • PRODUCT • ENGINEERING

MADE BY: ADITI NEVE

email: AditiNeve.csed25@adaniuni.ac.in



The Business Challenge

Problem

Enterprise customers are manually exporting dashboard data every week – a time-consuming, error-prone process that degrades experience.

Churn Risk

Several strategic accounts have flagged churn unless we provide automated, customized reporting – revenue at risk.

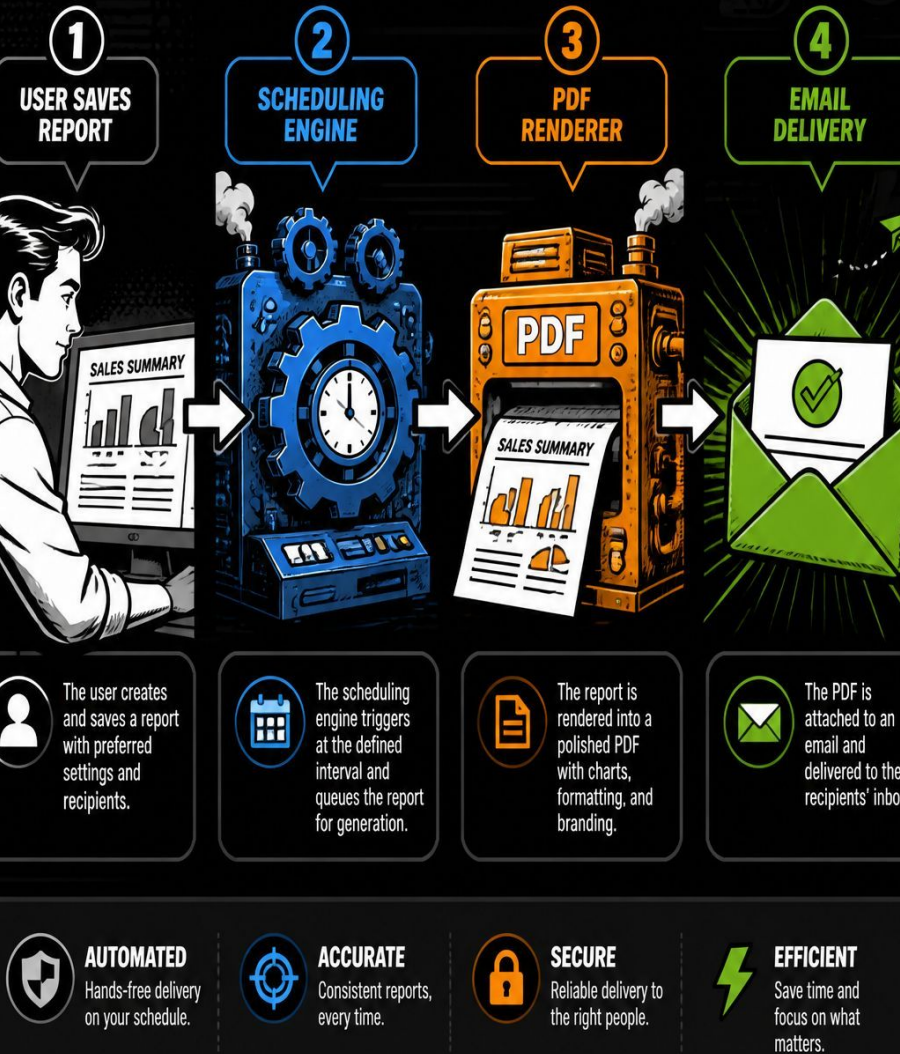
Opportunity

Deliver foundational BI features to improve stickiness, lower operational friction, and protect account retention.



REPORT DELIVERY WORKFLOW

— FROM SAVE TO INBOX —



Strategic Solution Overview

We propose a two-part platform enhancement: (1) Customizable Reporting to capture saved configurations and (2) Automated PDF Scheduling to deliver those saved reports on a recurring cadence.

- Scheduling depends on saved report configurations – implement reporting foundation first (Phase 1) before scheduling (Phase 2).

Core Capabilities & Dependencies

01

Customizable Reporting (Foundation)

Allow clients to define metrics, dimensions, date ranges, filters, and save configurations.

02

Automated PDF Scheduling (Delivery)

Recurring delivery of saved reports as PDFs with multi-recipient support and customizable email copy.

03

Core Dependency

Scheduling engine requires saved report objects created in Reporting (Phase 1 → Phase 2).

The screenshot displays a software interface for designing a deck, titled "THILSIDE DECK". The interface is organized into several panels:

- 1 SELECT CATEGORY:** A sidebar menu with icons for DECK STRUCTURE, STAIRS & RAILINGS, FOUNDATION, FRAMING, FASTENERS, MATERIALS, LOADS, and ENVIRONMENT.
- 2 DIMENSIONS & CONFIGURATION:** A 3D isometric view of a deck with dimension callouts: DECK LENGTH 5400 mm, DECK WIDTH 3600 mm, RAILING HEIGHT 1100 mm, HEIGHT FROM GRADE 1500 mm, and POST SPACING 1800 mm. Below the view are checkboxes for DIMENSIONS, POSTS, and FOOTINGS.
- 3 METRICS OVERVIEW:** A summary of key metrics:
 - DEAD LOAD: 2.75 kN (2.75 kN/m²)
 - LIVE LOAD: 4.79 kN (4.79 kN/m²)
 - SAFETY FACTOR: 2.35 (PASS)
 - MAX DEFLECTION: 8.6 mm (L/628)
- 4 MATERIAL SUMMARY:** A list of materials and their quantities:
 - DECKING (30x140): 18.9 m²
 - JOISTS (45x195): 31.0 m
 - DEAMS (90x195): 10.8 m
 - POSTS (90x90): 12.0 m³
 - CONCRETE: 0.42 m³
 - FASTENERS: 128 pcs
 - EST. WEIGHT: 324.7 kg
- 5 DIMENSION INPUTS:** A panel with sliders for adjusting various dimensions: DECK LENGTH (5400 mm), DECK WIDTH (3600 mm), HEIGHT FROM GRADE (1500 mm), POST SPACING (1800 mm), JOIST SPACING (400 mm), RAILING HEIGHT (1100 mm), and STAIR WIDTH (900 mm).
- 6 LOAD CONFIGURATION:** A panel for selecting load types and values:
 - LIVE LOAD (OCCUPANCY): Residential (4.79 kN/m²)
 - DEAD LOAD (INCLUDES SELF WEIGHT): Standard (2.75 kN/m²)
 - SNOW LOAD: Region 2 (1.25 kN/m²)
 - WIND LOAD: Basic Wind Speed 30 m/s (0.73 kN/m²)
- 7 PERFORMANCE DIAGRAMS:** A graph showing DEFLECTION (mm) vs Span (mm) with a curve peaking at 8.6 mm. A "COMPLIANT" stamp is visible.
- 8 CODE & REFERENCES:** A list of applicable standards: AS/NZS 1170.0, AS/NZS 1170.1, AS 1684, and NCC 2022.

Phased Delivery Strategy

Phase 1 — MVP: Reporting

Enable saved views, metric & dimension selection, basic filters, and persistence to stop immediate churn.

Phase 2 — MVP: Scheduling

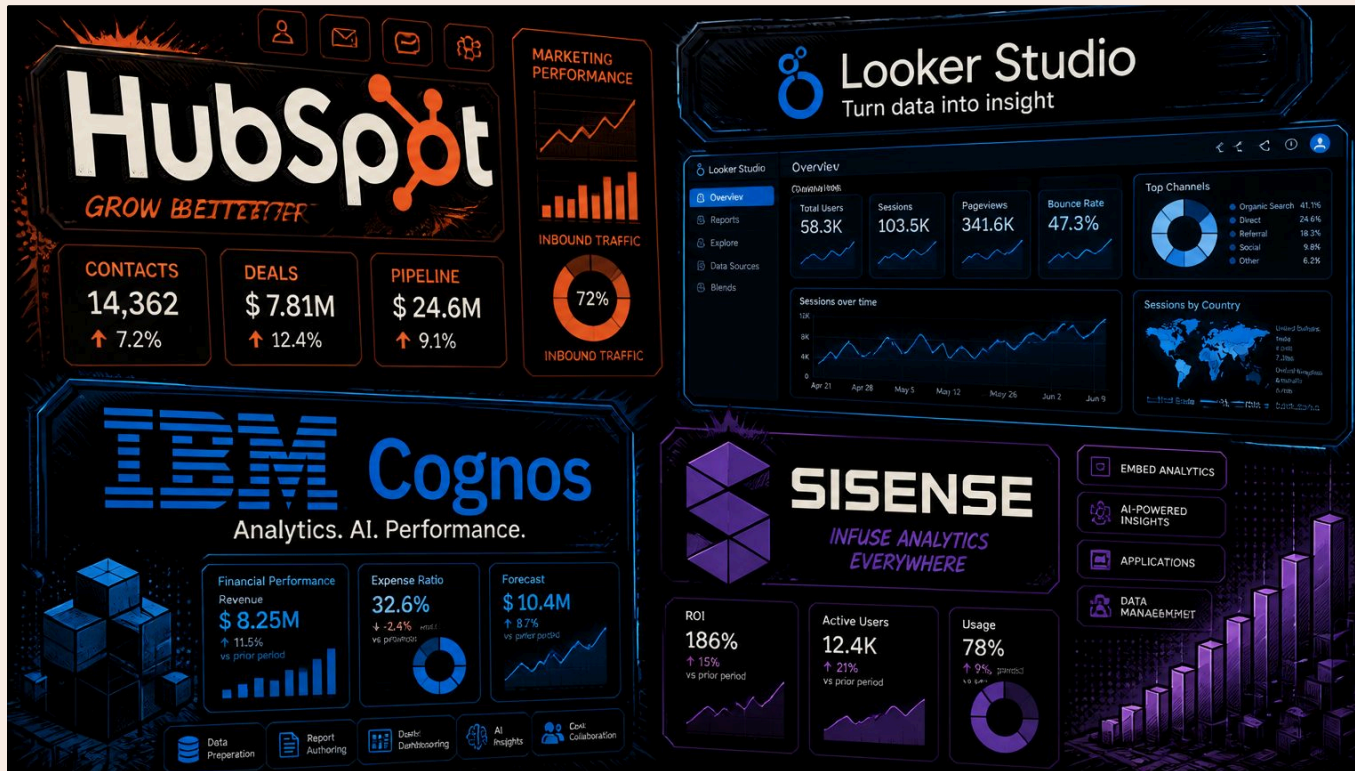
Implement recurring PDF rendering, multi-recipient email delivery, and schedule management UX.

Trade-offs

Defer advanced analytics, complex formulas, drag-and-drop builders, and white-labeling to post-MVP phases.



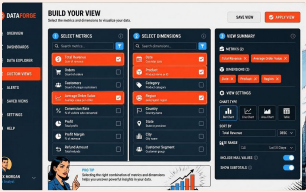
Market Benchmarking & Standards



We evaluated leading products to define baseline expectations:

- Custom reporting: flexible metric/dimension selection, saved views, single-level filters.
- Delivery: cron-style recurrence (daily/weekly/monthly), PDF attachments, customizable email subject/body.
- Parity goal: match core UX and delivery options that enterprise customers expect.

Product Requirements — MVP Scope



Reporting Engine

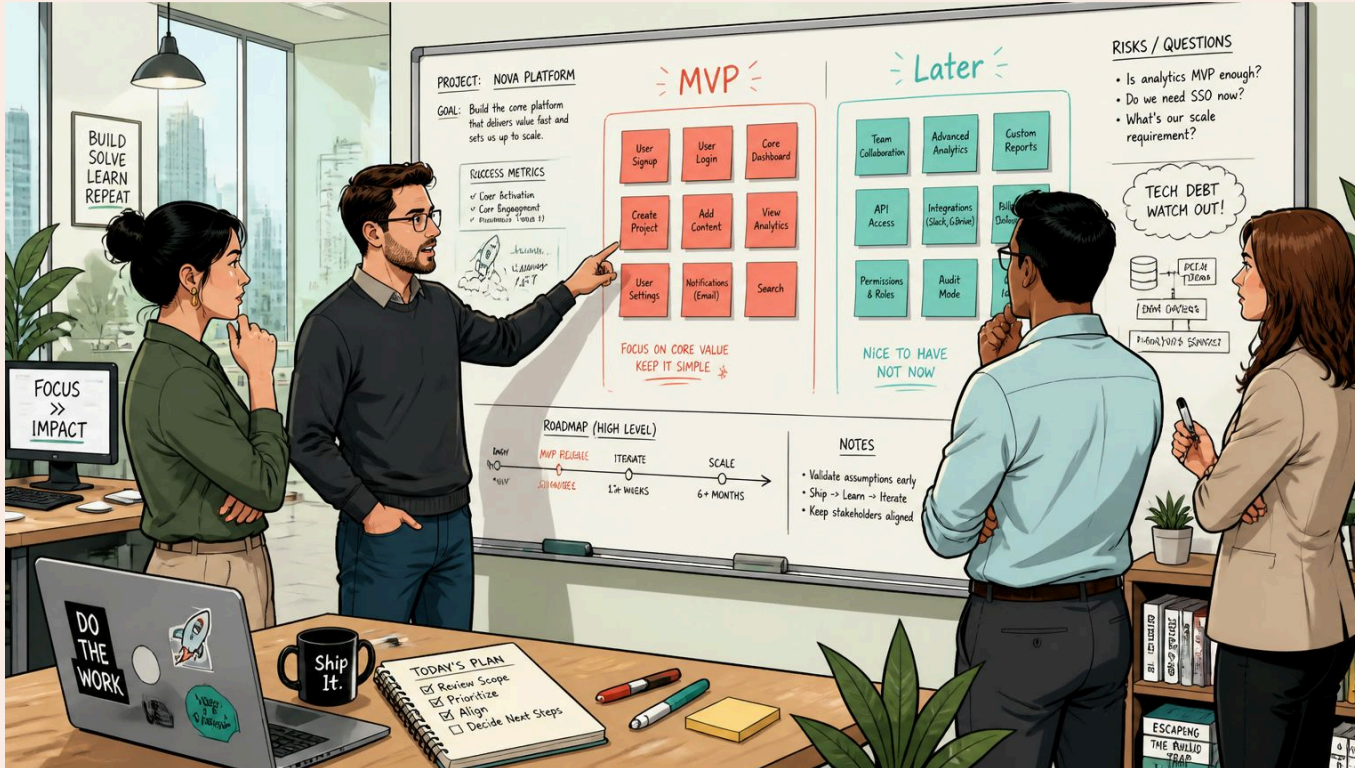
Metric & dimension pickers, presets and calendar date selectors, plus save/load/edit/delete report configs.



Scheduling Engine

Select a saved report, set recurrence (e.g., Every Monday 08:00), add multiple recipients, and queue background PDF rendering.

Scope Boundaries — Out of Scope (MVP)



Reporting Exclusions

Drag-and-drop builders, complex geospatial maps, custom calculated formulas, cross-object joins.

Scheduling Exclusions

Non-PDF exports (Excel/Slides), pre-delivery approval workflows, Slack/Teams integrations.

- ❑ Keeping MVP focused preserves engineering bandwidth and accelerates time-to-value.

Execution Roadmap & Agile Backlog

1 Reporting Stories

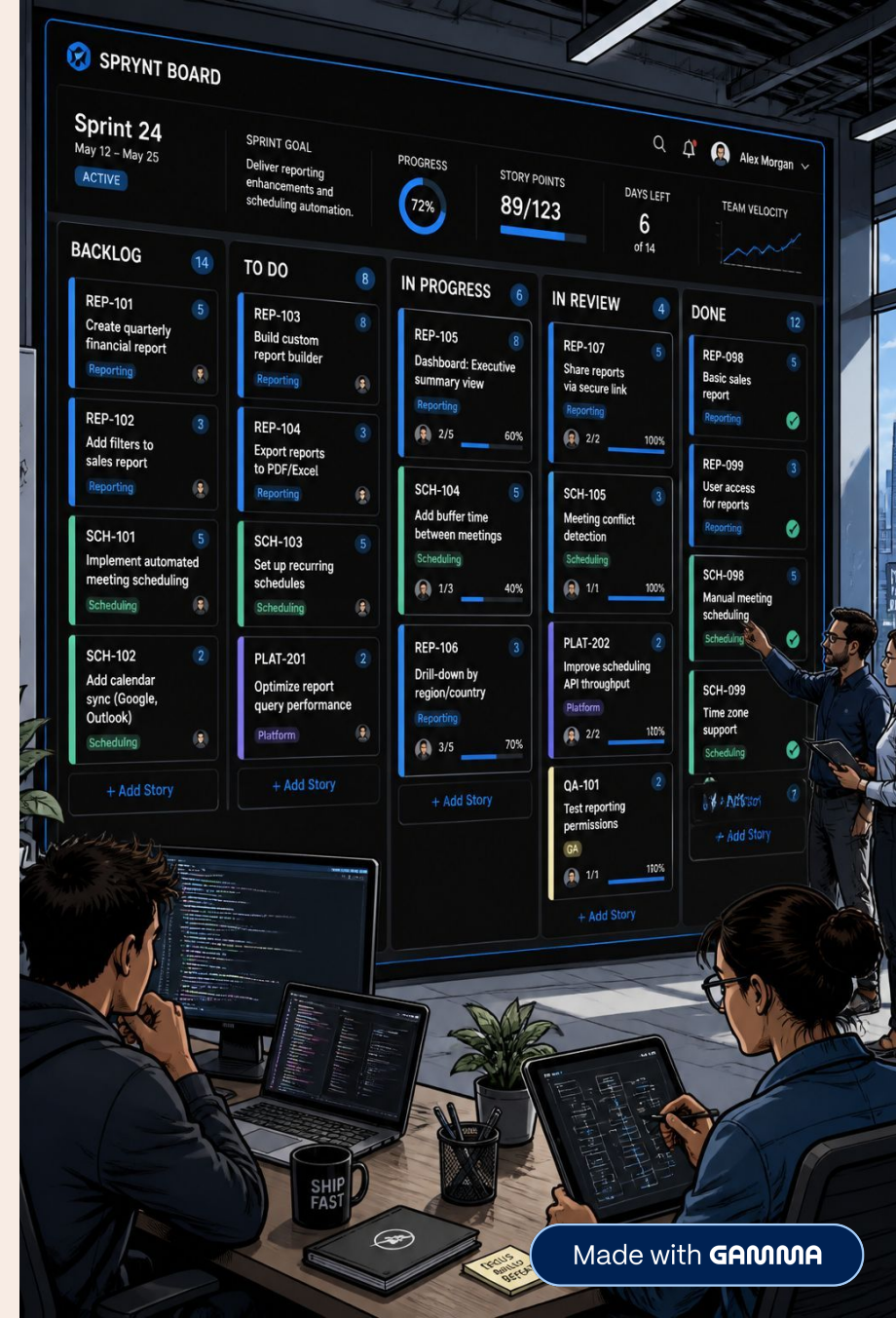
Create new custom report & select metrics (3 pts); Dimension breakdown & date picker (3 pts); Apply filter & save view (3 pts / 2 pts).

2 Scheduling Stories

Schedule recurring saved report (3 pts); Deliver PDF attachment engine (5 pts); Manage schedules interface (3 pts).

3 Delivery Plan

Plan sprints to complete Phase 1 first, then enable Phase 2 in subsequent sprint pairs. Prioritize reliability and monitoring.



Technical Considerations & Reliability

Async Job Queue

DB Cache

Idempotent Email

Next steps: finalize acceptance criteria for MVP stories, estimate sprint cadence, and schedule stakeholder demos after Phase 1 completion.

Design priorities that ensure scale and reliability.



Asynchronous Processing

Use background job queues (e.g., Celery/Redis) for PDF rendering and email tasks to avoid blocking request paths.



Database Caching

Cache aggregated query results to prevent Monday morning CPU spikes and speed report generation.



Idempotency & Retries

Implement idempotent delivery tokens and retry policies to avoid duplicate emails and handle transient failures.